

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

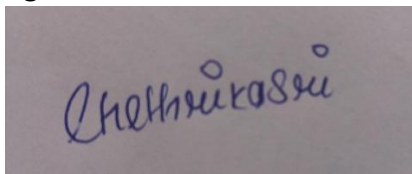
APPLICATION- BASED QUESTIONS

Code of Conduct:

I _____chethrikasri(192511112)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'Chethrikasri' written in a cursive style.

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.
 - a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
 - b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.
2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
 - a. Deduce the most likely original message.
 - b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.
3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.
 - a. Demonstrate how the encryption process transforms the message step-by-step.
 - b. Explain how the repeating keyword influences each character of the ciphertext.
4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
 - a. Analyze how investigators could detect the presence of hidden data within the image.
 - b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.
5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEEFEAOCAIVDEN”.
 - a. Determine the most probable original message by reversing the encryption process.
 - b. Critically evaluate why such a method is vulnerable in modern communication systems.
 - c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

TOPIC 3: SECURE MESSAGING SYSTEM USING VIGENÈRE CIPHER

Introduction

In today's digital era, secure communication is essential for protecting sensitive information from unauthorized access. Every day, personal, business, and government organizations exchange confidential data over the internet. Without proper security, this information can be intercepted, modified, or stolen by attackers. To prevent such threats, cryptography is used to ensure secure communication.

Cryptography is the science of converting readable information, called **plaintext**, into an unreadable format known as **ciphertext** using mathematical techniques. Only authorized users possessing the correct key can decrypt the ciphertext back into its original form. Cryptography plays a vital role in ensuring **confidentiality, integrity, authentication, and non-repudiation** of digital data.

One of the earliest and most important classical encryption techniques is the **Vigenère Cipher**. It is a **polyalphabetic substitution cipher** invented by Blaise de Vigenère in the 16th century. Unlike the Caesar Cipher, which shifts every letter by the same amount, the Vigenère Cipher uses a repeating keyword to determine different shift values for each letter. This significantly improves security by making the encrypted message more difficult to analyze.

Although modern encryption algorithms such as AES and RSA are used today, the Vigenère Cipher remains an important concept for understanding the fundamental principles of cryptography and secure communication.

Objectives

- To study the Vigenère Cipher.
- To understand the encryption and decryption processes.
- To demonstrate encryption using the keyword **KEY**.
- To analyze the importance of repeating keywords.
- To understand the advantages and limitations of classical cryptography.

THEORY AND WORKING PRINCIPLE

Basic Terminology

Plaintext: The original readable message before encryption.

Ciphertext: The encrypted message that cannot be understood without the correct key.

Encryption: The process of converting plaintext into ciphertext.

Decryption: The reverse process of converting ciphertext back into plaintext.

Keyword: A secret word used for encrypting and decrypting the message.

Working Principle of the Vigenère Cipher

The Vigenère Cipher encrypts each letter of the plaintext using the corresponding letter of a repeating keyword. Each letter of the keyword represents a different shift value. Therefore, the same plaintext letters may be encrypted differently depending on their positions.

Encryption Formula

$$C_i = (P_i + K_i) \bmod 26$$

Where:

- C_i = Ciphertext
- P_i = Plaintext
- K_i = Keyword

Decryption Formula

$$P_i = (C_i - K_i + 26) \bmod 26$$

Encryption Steps

- Write the plaintext.
- Repeat the keyword until it matches the message length.
- Convert letters into numbers ($A = 0, B = 1, \dots, Z = 25$).
- Add the plaintext and keyword values.

ENCRYPTION OF THE GIVEN MESSAGE

- **Assessment Question**
- A company uses the Vigenère Cipher to protect internal communication.
- Plaintext = **HELLO**
- Keyword = **KEY**
- **Step 1 – Repeat the Keyword**

Plaintext H E L L O

Keyword K E Y K E

- **Step 2 – Convert Letters into Numbers**

Letter	H	E	L	L	O
Value	7	4	11	11	14
Letter	K	E	Y	K	E
Value	10	4	24	10	4

- **Step 3 – Perform Encryption**

Plain	7	4	11	11	14
Key	10	4	24	10	4
Sum	17	8	35	21	18
Modulo 26	17	8	9	21	18
Cipher	R	I	J	V	S

- **Encrypted Message**
- **HELLO → RIJVS**
- The ciphertext generated after encryption is **RIJVS**.

DECRYPTION AND ANALYSIS

Decryption

Ciphertext = **RIJVS**

Keyword = **KEYKE**

Cipher	R	I	J	V	S
Value	17	8	9	21	18
Key	10	4	24	10	4
Subtract	7	4	-15	11	14
Add 26 (if negative)	7	4	11	11	14
Plaintext	H	E	L	L	O

The original message obtained after decryption is **HELLO**.

Influence of the Repeating Keyword

The keyword KEY is repeated as KEYKE to match the length of the plaintext. Each letter of the keyword provides a different shift value, causing each plaintext letter to be encrypted differently. As a result, repeated letters in the plaintext do not always produce the same ciphertext letters.

This characteristic makes the Vigenère Cipher stronger than the Caesar Cipher because it reduces predictable patterns and makes frequency analysis more difficult. However, if a short keyword is repeated many times, attackers can still perform cryptanalysis using techniques such as the Kasiski Examination or Friedman Test. Therefore, using a long and random keyword significantly improves security.

ADVANTAGES, LIMITATIONS AND APPLICATIONS

Advantages

- Provides better security than the Caesar Cipher.
- Uses multiple substitution alphabets.
- Makes frequency analysis more difficult.
- Simple and easy to implement.
- Requires minimal computational resources.
- Useful for understanding the principles of cryptography.

Limitations

- Repeating keywords reduce security.
- Can be broken using Kasiski Examination.
- Not suitable for modern secure communication.
- Secure key distribution is required.

Applications

- Educational demonstrations of cryptography.
- Historical military communications.
- Coding competitions and puzzles.
- Learning encryption and decryption techniques.
- Foundation for understanding modern encryption algorithms.

PAGE 6 – COMPARISON, CONCLUSION AND REFERENCES

Comparison

Feature	Caesar Cipher	Vigenère Cipher	AES
Cipher Type	Monoalphabetic	Polyalphabetic	Block Cipher
Key	Single Shift	Keyword	128/192/256-bit Key
Security	Low	Medium	Very High
Modern Usage	No	Educational	Yes

Conclusion

The Vigenère Cipher is one of the most significant classical encryption techniques because it improves security by using a repeating keyword to generate different shift values for each letter of the plaintext. In this assignment, the message **HELLO** was successfully encrypted using the keyword **KEY**, resulting in the ciphertext **RIJVS**, and then correctly decrypted back to the original message.

Although the Vigenère Cipher is not secure enough for modern applications, it remains an excellent learning tool for understanding encryption, key management, and secure communication. It also forms the foundation for studying advanced cryptographic algorithms used in today's cybersecurity systems.

References

1. William Stallings, *Cryptography and Network Security: Principles and Practice*, Pearson.
2. Behrouz A. Forouzan, *Cryptography and Network Security*, McGraw-Hill.
3. National Institute of Standards and Technology (NIST) – Cryptographic Standards.
4. Lecture Notes on Classical Cryptography.

For higher marks: Draw the **Vigenère Table (Tabula Recta)** and the **Encryption/Decryption flow diagram** neatly in your assignment. These diagrams enhance presentation and demonstrate a clear understanding of the topic.

TOPIC 4: SUSPICIOUS DIGITAL ASSET INVESTIGATION USING STEGANOGRAPHY

Introduction

In today's digital world, images, videos, audio files, and documents are exchanged frequently through emails, social media, and cloud platforms. While encryption protects the content of a message, it also reveals that a secret message exists. In contrast, **steganography** hides the very existence of the message by embedding it within a digital file, making it invisible to others.

Steganography is widely used for secure communication, digital watermarking, copyright protection, and military applications. However, cybercriminals may misuse this technique to secretly transmit confidential information, malware, or illegal data without attracting attention. Therefore, digital investigators and cybersecurity professionals must understand both the methods used to hide information and the techniques used to detect it.

Application Scenario

An investigator suspects that an employee has hidden confidential company information inside a digital image before sharing it online. The investigator's objective is to determine whether the image contains hidden data, identify the hiding technique used, recover the concealed information, and recommend methods to improve the security of steganographic communication.

Objectives

- To understand the concept of steganography.
- To explain how information is hidden inside digital images.
- To discuss techniques used by investigators to detect hidden data.
- To suggest methods for improving the security of steganographic communication.

FUNDAMENTALS OF STEGANOGRAPHY

What is Steganography?

Steganography is the practice of hiding secret information inside another digital file, known as the **cover medium**, so that the existence of the message remains unknown. Unlike encryption, which converts data into unreadable text, steganography conceals the message within a normal-looking file.

Components of Steganography

- **Cover Image:** The original image used to hide data.
- **Secret Message:** The confidential information to be hidden.
- **Stego Key:** A password or key used during embedding and extraction.
- **Stego Image:** The final image containing the hidden information.

Types of Steganography

- **Image Steganography** – Hides data inside images.
- **Audio Steganography** – Embeds data in sound files.
- **Video Steganography** – Conceals information in video frames.
- **Text Steganography** – Hides data within text documents.

Among these, **image steganography** is the most commonly used because digital images contain millions of pixels, providing enough space to hide information without noticeable changes.

HOW INFORMATION IS HIDDEN INSIDE AN IMAGE

Least Significant Bit (LSB) Technique

The **Least Significant Bit (LSB)** technique is one of the simplest and most popular methods of image steganography.

Each pixel in a digital image is represented by binary values. The least significant bit of selected pixels is replaced with bits of the secret message. Since only the last bit changes, the modification is almost impossible for the human eye to detect.

Example

Original Pixel:

11001010

Secret Bit:

1

Modified Pixel:

11001011

Only the last bit changes, but the image appears visually unchanged.

Working Process

1. Select a cover image.
2. Convert the secret message into binary.
3. Replace the least significant bits of image pixels with secret bits.
4. Save the modified image as the **Stego Image**.
5. At the receiver's end, use the same algorithm and key to extract the hidden message.

DETECTING HIDDEN INFORMATION (STEGANALYSIS)

Techniques Used by Investigators

Investigators use **steganalysis** to determine whether a digital image contains hidden information.

1. Histogram Analysis

Compares the pixel distribution of an image to detect unusual changes caused by hidden data.

2. Metadata Analysis

Examines image properties such as file size, creation date, editing software, and compression details for suspicious modifications.

3. Statistical Analysis

Uses mathematical models to identify irregularities in pixel values that may indicate embedded information.

4. File Size Comparison

An image containing hidden data may have a slightly different file size than the original.

5. Pixel Analysis

Compares neighbouring pixels to identify abnormal patterns created during data embedding.

6. Steganalysis Tools

Investigators use specialized software to automatically scan images for hidden information and recover concealed messages.

These methods help digital forensic experts determine whether confidential data has been secretly transmitted.

IMPROVING THE SECURITY OF STEGANOGRAPHIC COMMUNICATION

To make steganographic communication more secure, the following methods should be adopted:

Encrypt the Message Before Hiding

The secret message should first be encrypted using strong algorithms such as **AES** before embedding it into an image.

Use Strong Stego Keys

A complex password should be used to prevent unauthorized extraction.

Random Pixel Embedding

Instead of embedding data sequentially, random pixel locations should be selected to reduce detection.

Large Cover Images

High-resolution images provide more space for hiding data while minimizing visible changes.

Compression Resistance

Choose embedding techniques that preserve hidden information even after image compression.

Multi-Layer Security

Combining encryption with steganography provides two levels of protection, making attacks significantly more difficult.

APPLICATIONS, ADVANTAGES, LIMITATIONS, CONCLUSION AND REFERENCES

Applications

- Secure military communication.
- Digital watermarking for copyright protection.
- Secure transmission of confidential business documents.
- Protection of medical records.
- Cybersecurity and digital forensics.

Advantages

- Conceals the existence of confidential information.
- Difficult to detect visually.
- Enhances privacy during communication.
- Can be combined with encryption for stronger security.

Limitations

- Large hidden messages may reduce image quality.
- Advanced steganalysis tools can detect hidden data.
- File modifications or compression may destroy hidden information.

Conclusion

Steganography is an effective technique for concealing confidential information inside digital images without revealing its existence. In the given investigation scenario, forensic experts

analyze the suspicious image using steganalysis techniques such as histogram analysis, metadata examination, and statistical analysis to detect hidden data. By combining **encryption** with **steganography**, organizations can significantly improve the confidentiality and security of sensitive communications while reducing the risk of unauthorized access.

References

1. William Stallings, *Cryptography and Network Security: Principles and Practice*, Pearson.
2. Behrouz A. Forouzan, *Cryptography and Network Security*, McGraw-Hill.
3. Neil F. Johnson, Zoran Duric, and Sushil Jajodia, *Information Hiding: Steganography and Watermarking*.
4. National Institute of Standards and Technology (NIST), Digital Forensics Guidelines.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation